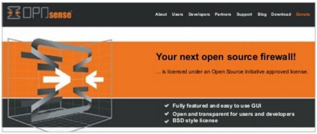

Figure 1: OPNsense, a free and powerful open source firewall solution

as well as a routing platform for network security and cyber forensic investigation. It is free, open source and is available under the FreeBSD licence. It has excellent features to guard the network against assorted attacks and malicious intrusions.

OPNsense has most of the secured modules and features that are normally only available with the very expensive proprietary and commercial firewall systems. It has the Bootstrap Framework based GUI that has a Web based interface to enable and deliver the easy-to-use platform without any complexities.

The core powerful features of OPNsense include:

- Forward caching proxy and blacklisting
- Rule based traffic transmission
- Traffic shaper
- Supports IP-less firewalls

You may refer to the website and the various resources available on the Internet for more features.

Installation and working with OPNsense

A USB memstick installer can be used. Simply write the downloaded image to a USB drive or hard disk and the steps shown below can be followed.

For FreeBSD, type:

```
dd if=OPNsenseVERSION of=/dev/daX bs=16k
```

For Linux, the command is:

```
dd  if=OPNsenseVERSION of=/dev/sd(IDE-Device) bs=16k
```

For OpenBSD, type:

```
dd if=OPNsenseVERSION of=/dev/rsd6c bs=16k
```

The command for Mac OS X is:

```
sudo dd  if=OPNsense img of=/dev/<RawDevice>disk<DiskDeviceNumber> bs=64k
```

For Windows, type:

```
physdiskwrite -u OPNsenseVERSION
```

Figure 2: SSL integration with OPNsense

Figure 3: User interface of OPNsense

Integration with OpenSSL / LibreSSL

The OPNsense images are integrated with OpenSSL and can be selected on demand. LibreSSL is usable and selected from the GUI as *System -> Settings -> General.*

The default user name in OPNsense is *root* and the password is *opnsense.*

Key components and the GUI of OPNsense

Lobby: Click on the OPNsense logo in the interface and navigate the Lobby as well as the dashboard directly.

From OPNsense Lobby, the following options can be selected:

- Dashboard
- Widgets
- Licence information
- Change password credentials
- Logout

Menu: From Menu, the network administrator can select the various security and configuration features to be changed and customised.

There are three levels of layers in Menu:

- Category level
- Function level
- Configuration level

The depiction shown in Figure 4 directs the user to change the settings and then select from the firewall.

Settings -> General

Setting up the IDS and intrusion prevention system

To enforce and enable the IDS/IPS, select *Services -> Intrusion Detection* and then enable the IPS mode.